

Hearsay

A 3-minute hearing-in-noise check that finds the family member quietly saying "what?" — UnivaBio 2026 · AI for Human Health · Sharon Basovich (solo)

Problem

Understanding speech in noise is often the first hearing ability to slip, and people often wait years before acting. Online hearing tests usually skip speech-in-noise, and they can't know how loud your headphones are — yet many still report a "normal" range.

Solution

- **Digits-in-noise check:** hear 3 spoken digits in speech-shaped noise, type them; antiphasic presentation; a headphone check first.
- **Bayesian adaptive engine** (the AI): posterior over threshold $\times$ slope; each next noise level minimizes expected posterior entropy; stops when the 90% credible interval is tight (~ 12 triplets).
- **Same-device family board:** compares each person with the median of others on the same device, so calibration offsets cancel. Flags only gaps > 3 dB *and* beyond measurement uncertainty.
- **Hear-through simulator** and a reproducible in-browser **Lab**.

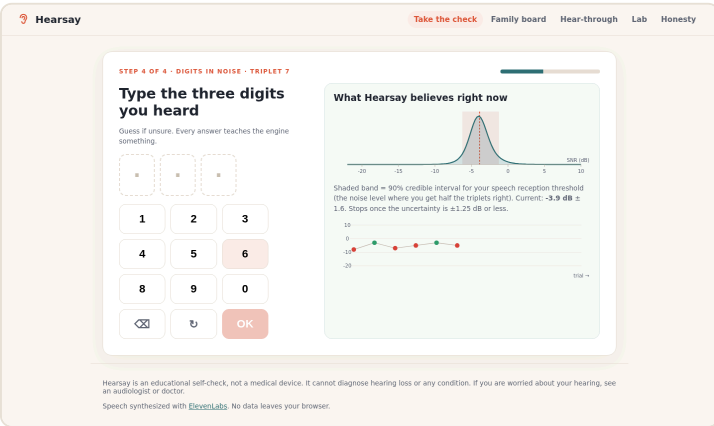
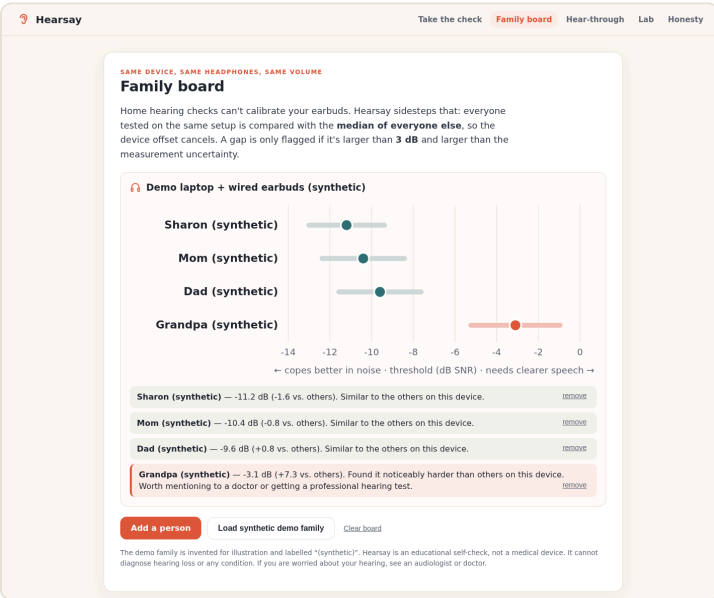
Audience & impact

Families and caregivers of older adults, and students exposed to loud music. The output is actionable and non-alarming: "found it noticeably harder than others on this device — worth mentioning to a doctor."

Measured results (simulated listeners, n=400, tested twice)

Procedure	Triplets	RMSE	90% CI coverage
Staircase, 24 trials	24	0.79 dB	62%
Staircase, 12 trials	12	1.53 dB	54%
Hearsay Bayesian	11.8	1.01 dB	94%

$\sim 34\%$ lower error than a staircase of the same length; half the trials of the classic track, which remains slightly more precise.



Built

TypeScript, Vite, Web Audio API, hand-written FFT, Vitest (32 tests). No backend; results stay in the browser.

Honesty

Not a medical device; not a diagnosis; not clinically validated. Validation uses simulated listeners only. Demo family is invented and labelled. Speech synthesized once, offline, with ElevenLabs TTS; noise generated locally; live site calls no APIs. Built with the Devin AI coding agent, directed and reviewed by Sharon. No human-subject data. Planned next: ethics-approved pilot against a calibrated clinical test.